MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
MAKE-UP EXAMINATION		**DEC, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**FC**
Course Name	**Design of prestressed concrete structures**	**Course Code**	**CIE 3124**
Semester	**V**	**Max Marks**	**50**
Date	**03-01-2025**	**Duration**	**3 hrs**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Explain the following i) Load balancing concept ii) Hoyer effect	**4**	1	1,2,8,12	**2**
1B	A simply supported fully pre-tensioned PSC beam of span 15m is pre-stressed with a parabolic cable having zero eccentricity at supports and eccentricity of 100mm below the center of gravity of section (C.G.C.) at mid-span. The initial pre-stressing force induced is 500 kN with 20% loss of prestress . The cross section of beam is an unsymmetrical I-section with top flange 450mm x 100 mm, web 100mm x 600mm and bottom flange 350mm x 200mm, with I = 1.79 x 10^{10}mm^4 and y_t = 470mm. If the beam is subjected to a working load of 10 kN/m throughout the span, Illustrate the stress distribution diagram across the depth at mid-span for the transfer and working stage.	**6**	2	1,2,3,8,12	**3**
2A	A PSC beam designed to carry a live load of 15 kN/m over a span of 12 m. Grade of Concrete: M-35; Characteristic Strength of Steel: 1700 N/mm^2; Area of steel: 500mm^2; Effective pre-stressing force: 650kN; Cable profile: Parabolic, which is concentric at supports and having an eccentricity of 250mm at mid-span; Effective cover to longitudinal	**5**	3	1,2,12	**3**

	reinforcement: 100mm. Determine the ultimate shear resistance of the section uncracked in flexure and provide shear reinforcement at support using 8mm diameter HYSD bars. Take the Ultimate shear resistance of the section cracked in flexure as 100kN. Beam Cross section details are as follows. Top flange: 450mm × 150mm, Web: 150mm × 630mm, Bottom flange: 200mm×300mm, y_t = 472.09mm, I = 2.69×10^{10} mm^4.				
2B	A prestressing force of 400KN is transmitted through a Fressynet anchor of diameter 135mm, which is located at the center of the end block of section 200mm × 350mm. For M-35 grade of concrete and Fe 415 grade steel, examine the bearing stress and Illustrate the end block design.	**5**	4	1,2,12	**3**
3A	A PSC beam designed to carry a live load of 15 kN/m over a span of 12 m. Grade of Concrete: M-35; Characteristic Strength of Steel: 1700 N/mm^2; Area of steel: 500mm^2; Effective pre-stressing force: 650kN; Cable profile: Parabolic which is concentric at supports and having an eccentricity of 250mm at mid span. Effective cover to longitudinal reinforcement: 100mm. Examine the post tensioned beam for limit state of collapse in flexure for the beam cross section details as follows. Top flange: 450mm × 150mm, Web: 150mm × 630mm, Bottom flange: 200mm × 300mm, y_t = 472.09mm, I = 2.69×10^{10} mm^4 .	**6**	3	1,2,12	**3**
3B	Discuss the advantages of Composite Construction in Prestressed Concrete structures.	**4**	5	1,2,12	**2**
4A	A simply supported unbonded post tensioned PSC beam of span 12m is prestressed with two cables induced with 600kN prestress in each cable. Cable 1 is parabolic having zero eccentricity at supports and an eccentricity of 100mm below center of gravity of section (C.G.C.) at mid span. Cable 2 is a straight cable having eccentricity of 75mm above centroid throughout the span. The beam has an unsymetrical I section with top flange 450mm x 120 mm, web 120mm x 600mm and bottom flange 250mm x 200mm, with I = 1.74 x $10^{10}mm^4$. The beam is subjected to a working load of 15 kN/m throughout the span. Taking loss of prestress as 20%, and creep coefficient as 1.5 for M-35 grade concrete, examine for limit state of serviceability at midspan as per the codal provisions.	**6**	3	1,2,12	**3**

4B	A simply supported pre-tensioned beam of span 13m having an overall depth of 1080mm is loaded with a working load of 17kN/m. The beam is prestressed with 15 numbers of 5mm diameter indented wires with an initial prestress of 1200 N/ mm^2 and an effective prestress of 1000 N/mm^2. The stress in the wires at critical section is 1200 N/mm^2. The effective cover to the centroid of pre-stressing steel is 275mm, and the bond strength for the M40 grade of concrete is 1.9MPa. Examine for development length and end zone reinforcement.	**4**	4	1,2,12	**3**
5A	Analyze to design a simply supported type-1 pre-tensioned PSC beam of span 11m loaded with a working load of 18kN/m. Take initial pre-stressing force as 1500kN, loss of pre-stress as 18%, M40 grade of concrete, and trial cross-section as follows: Top flange: 600mm × 250mm, Web:250mm × 600mm, Bottom flange: 300mm × 300mm, with y_t=490.38mm, I = 4.99×10^{10} mm^4.	**8**	2	1,2,3,8,12	**4**
5B	Explain the transmission zone and its significance in pre-tensioned prestressed members.	**2**	4	1,2,12	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***4***
2	***14***
3	***17***
4	***11***
5	***4***

BL	Total Marks
2	**7**
3	**38**
4	**8**
5	